

Project Requirement Analysis Phase

Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-5694
Team Size	4
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Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Technical Architecture

S.No	Component	Description	Technology
1	User Interface	Web interface through which users access ServiceNow Service Catalog and request status.	ServiceNow Web UI
2	Application Logic-1	Flow-based automation that processes the approved Standard Laptop request.	ServiceNow Flow Designer
3	Application Logic-2	Action that creates a Catalog Task and sets task values.	Create Catalog Task action
4	Request / Task Data	Stores the Requested Item and Catalog Task information used by the workflow.	ServiceNow Service Catalog / Catalog Task
5	Approval	Provides the approval state used before the automated task is created.	ServiceNow approval mechanism
6	Assignment	Routes the configuration task to the responsible team.	Hardware assignment group
7	External API	No external API is required for the described workflow.	Not applicable
8	Machine Learning Model	No ML model is required for standard laptop procurement automation.	Not applicable
9	Infrastructure	Application, Flow Designer and Service Catalog execute within the ServiceNow platform.	ServiceNow instance / platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source framework is required; the solution uses ServiceNow platform capabilities.	ServiceNow Flow Designer / Service Catalog
2	Security Implementations	The workflow follows the approval state before creating the configuration task and uses controlled task assignment.	ServiceNow workflow and approval
3	Scalable Architecture	The flow can be extended to other standard procurement items and task-routing requirements.	ServiceNow Flow Designer
4	Availability	The automation operates when the ServiceNow instance and required request data are available.	ServiceNow instance

S.No	Characteristics	Description	Technology
5	Performance	Use a focused trigger and action sequence to avoid unnecessary workflow processing.	Flow Designer